

Python Basics

Question 1 : What is Python and why is it widely used in Data Analytics? Mention any three reasons.

Answer:

Python is a high-level, easy-to-learn programming language used for web development, automation, artificial intelligence, and data analytics. In Data Analytics, Python helps analyze, clean, and visualize data efficiently.

Three reasons why Python is widely used in Data Analytics:

1. **Easy Syntax** – Python has simple and readable syntax, making it beginner-friendly.
2. **Powerful Libraries** – Libraries like Pandas, NumPy, and Matplotlib help in data analysis and visualization.
3. **Large Community Support** – Python has a huge community and many online resources for learning and problem-solving.

Question 2 : Explain the difference between List and Tuple in Python.

Answer:

Difference Between List and Tuple in Python

List and Tuple are both used to store multiple values in Python, but they have some important differences :

List	Tuple
List is mutable (values can be changed).	Tuple is immutable (values cannot be changed).
Uses square brackets [].	Uses parentheses ().
Slower in execution.	Faster in execution.
More memory is used.	Less memory is used.
Suitable for changing data.	Suitable for fixed data.

Example: [1 , 2 , 3]	Example: (1 , 2 , 3)
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Question 3 : What is a function in Python? Why are functions useful?

Answer:

A function is a block of reusable code that performs a specific task. Functions help avoid repeating code and make programs easier to understand.

Benefits of Functions:

- Reusability
- Better organization
- Easy debugging
- Saves time

Example:

```
def greet():  
  
    print("Hello!")  
  
greet()
```

Question 4 : Write a Python program to take a user's name as input and print a greeting message.

Answer:

```
name = input("Enter your name: ")  
  
print("Hello,", name, "! Welcome!")
```

Question 5 : Write a Python program to check whether a number is even or odd using conditional statements.

Answer:

```
num = int(input("Enter a number: "))  
  
if num % 2 == 0:
```

```
        print("Even Number")
    else:
        print("Odd Number")
```

Question 6 : Write a program to print numbers from 1 to 10 using a loop.

Answer:

```
for i in range(1, 11):
    print(i)
```

Question 7 : Create a list of five numbers and print the maximum number from the list.

Answer:

```
numbers = [12, 45, 7, 89, 34]
print("Maximum number is:", max(numbers))
```

Question 8 : Write a Python program to remove duplicate values from a list using a set.

Answer:

```
numbers = [1, 2, 2, 3, 4, 4, 5]
unique_numbers = list(set(numbers))
print("List after removing duplicates:", unique_numbers)
```

Question 9 : Write a function that returns the square of a number.

Answer:

```
def square(num):
    return num * num
```

```
print(square(5))
```

Question 10 : Write a Python program to count how many times a number appears in a list. Example List [2,3,4,2,5,2]

Answer:

Example List:

```
[2,3,4,2,5,2]
```

Program:

```
numbers = [2, 3, 4, 2, 5, 2]
```

```
count = numbers.count(2)
```

```
print("2 appears", count, "times in the list")
```

Output:

```
2 appears 3 times in the list
```